

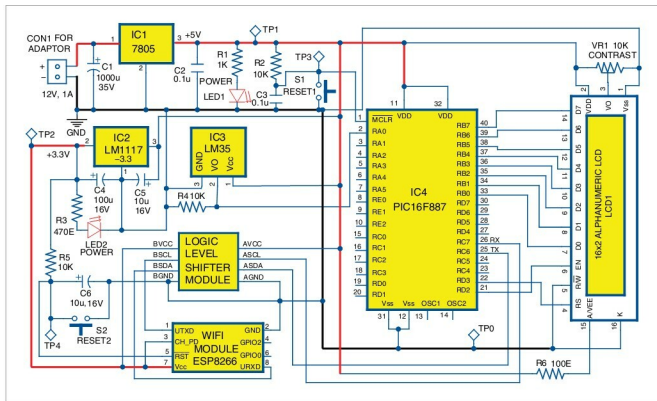


Fig. 3: Circuit diagram of the temperature-logging system

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - LM1117-3.3, 3.3V linear regulator
- IC3 - LM35 temperature sensor
- IC4 - PIC16F887 MCU
- LED1-LED2 - 5mm LED
- Level-shifter - 8-pin logic-level shifter
- Wi-Fi module - 8-pin ESP8266 Wi-Fi module
- LCD - 16x2 alphanumeric LCD

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2, R4, R5 - 10-kilo-ohm
- R3 - 470-ohm
- R6 - 100-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2, C3 - 0.1 μ F ceramic disk
- C4 - 100 μ F, 16V electrolytic
- C5, C6 - 10 μ F, 16V electrolytic

Miscellaneous:

- S1-S2 - Tactile switch
- CON1 - DC adaptor connector
- Adaptor - 12V, 1A DC power supply
- 8-pin berg strip female connector for level-shifter module
- 8-pin pin berg strip female connector for Wi-Fi module



Fig. 4: ESP8266 Wi-Fi module



Fig. 5: Logic-level shifter

5V to 3.3V. We used an external logic shifter/convertor as shown in Fig. 5. +3.3V, CH_PD and RST pins are kept at 3.3V for normal functioning of ESP8266. Logic-level convertor is a device that safely steps down

5V signal to 3.3V.

LM35 temperature sensor. The LM35 series is of precision integrated circuit temperature sensors, whose output voltage is linearly proportional to Celsius (Centigrade) temperature. LM35 is rated to operate over -55°C to $+150^{\circ}\text{C}$, while LM35C is rated for a -40°C to a $+110^{\circ}\text{C}$ temperature range. This sensor has a linear scale factor of $10\text{mV}/^{\circ}\text{C}$ and operates from 4V to 30V.

Power supply. Power supply section includes a 5V regulator 7805 (IC1) and a 3.3V regulator LM1117-3.3 (IC2). The MCU, logic-level shifter and LCD are powered by 5V and ESP Wi-Fi module by 3.3V supply. Voltage for ESP8266 for all its pins is 3.3V. If you are using an adjustable voltage regulator, carefully set proper voltage to 3.3V.

In this project, temperature data from LM35 sensor is fed to pin 2 of PIC16F887 MCU. This data is given to ESP8266 Wi-Fi module through a logic-level shifter. Temperature data